

**CLASS XIIth- Computer Science
CBSE-2019**

**Time: 3 hour
Max Marks :70**

General Instructions :

- (i) SECTION A refers to programming language C++.**
- (ii) SECTION B refers to programming language Python.**
- (iii) SECTION C is compulsory for all.**
- (iv) Answer either SECTION A or SECTION B.**
- (v) It is compulsory to mention on the page 1 in the answer-book whether you are attempting SECTION A or SECTION B.**
- (vi) All questions are compulsory within each section.**
- (vii) Questions 2(b), 2(d), 3 and 4 have internal choices.**

SECTION-A

Question1:

(a) Write the type of C++ Operators (Arithmetic, Logical, and Relational Operators) from the following: (2)

- (i) !**
- (ii) !=**
- (iii) &&**
- (iv) %**

Ans.

(i) Logical (ii) Relational (iii) Logical (iv) Arithmetic

(b) Observe the following program very carefully and write the name of those header file(s), which are essentially needed to compile and execute the following program successfully: (1)

```
void main()
{
char text[20], newText[20];
gets(text);
strcpy(newText, text);
for(int i=0; i<strlen(text); i++)
if(text[i] == 'A')
text[i] = text[i]+2;
puts(text);
}
```

Answer:

- stdio.h
- string.h

(c) Rewrite the following C++ code after removing any/all Syntactical Error(s) with each correction underlined. (2)

Note: Assume all required header files are already being included in the program.

```
#define float PI 3.14
void main( )
{
float R=4.5,H=1.5;
A=2*PI*R*H + 2*PI*pow(R,2);
cout<<'Area='<<A<<endl;
}
```

```
#define PI 3.14 //Error 1
void main( )
{
float R=4.5,H=1.5;
A=2*PI*R*H + 2*PI*pow(R,2);
cout<<'Area='<<A<<endl;
}
```

OR

(d) Find and write the output of the following C++ program code: (2)

Note: Assume all required header files are already being included in the program.

```
void main( )
{
int Ar[ ] = { 6 , 3 , 8 , 10 , 4 , 6 , 7 };
int *Ptr = Ar , l ;
cout<<++*Ptr++ << '@' ;
l = Ar[3] -Ar[2] ;
cout<<++*(Ptr+l)<<'@'<<"\n" ;
cout<<+l + *Ptr++ << '@' ;
cout<<*Ptr++ <<'@'<< '\n' ;
for( ; l >=0 ; l -=2)
cout<<Ar[l] << '@' ;
}
```

Answer:

7@11@
6@8@

11@3@

(e) Find and write the output of the following C++ program code:

(3)

```

Typedef char STRING[80];
void MIXNOW(STRING S)
{
    Int Size=strlen(S);
    for(intl=0;l<Size;l+=2)
    {
        Char WS=S[l];
        S[l]=S[l+1];
        S[l+1]=WS;
    }
    for (l=1;l<Size;l+=2)
    if (S[l]>='M' && S[l]<='U')
    S[l]='@';
    }
void main()
{
    STRING Word="CBSEEXAM2019";
    MIXNOW(Word);
    cout<<Word<<endl;
}

```

Ans.

BCE@XEMA0291

(f) Observe the following program and find out, which output(s) out of (i) to (iv) will be expected from the program? What will be the minimum and the maximum value assigned to the variable Alter? (2)

Note: Assume all required header files are already being included in the program.

```

void main( )
{
    randomize( );
    int Ar[]={10,7}, N;
    int Alte r=random(2) + 10 ;
    for (int C=0;C<2;C++)
    {
        N=random(2) ;
        cout<<Ar[N] +Alter<<"#";
    }
}

```

(i) 21#20# (ii) 20#18#

(iii) 20#17# (iv) 21#17#

Ans.

The output expected from the program is (iii) 20#17#

Minimum Value of Alter = 10

Maximum Value of Alter = 11

Question 2:

(a) What is a copy constructor? Illustrate with a suitable C++ example. (4)

(b) Write the output of the following C++ code. Also, write the name of feature of Object Oriented Programming used in the following program jointly illustrated by the Function 1 to Function 4. (2)

Ans.

(a)

A copy constructor is an overloaded constructor in which an object of the same class is passed as reference parameter.

```
class X
{
    int a;
public:
    Y()
    {
        p=0;
    }
    Y(Y &ob)    //copy constructor
    {
        p=ob.p;
    }
};
```

(b)

```
void My_fun ( )    // Function 1
{
    for (int l=1 ; l<=50 ; l++) cout<< "-";
    cout<<end1 ;
}
void My_fun (int N)    // Function 2
{
    for (int l=1 ; l<=N ; l++) cout<<"*";
    cout<<end1 ;
}
void My_fun (int A, int B) // Function 3
{
    for (int l=1. ;l<=B ;l++) cout <<A*I ;
    cout<<end1 ;
}
```

```

}
void My_fun (char T, int N)    // Function 4
{
    for (int l=1 ; l<=N ; l++) cout<<T ;
    cout<<endl;
}
void main ( )
{
    int X=7, Y=4, Z=3;
    char C='#' ;
    My_fun (C,Y) ;
    My_fun (X,Z) ;
}

```

OR

Write any four differences between Constructor and Destructor function with respect to object oriented programming (2)

Ans.

Constructor	Destructor
Name of the constructor function is same as that of class	Name of the destructor function is same as that of class preceded by ~
Constructor functions are called automatically at the time of creation of the object	Destructor functions are called automatically when the scope of the object gets over
Constructor can be overloaded	Destructor cannot be overloaded
Constructor is used to initialize the data members of the class	Destructor is used to de-initialize the data members of the class

(c) Define a class Ele_Bill in C++ with the following descriptions:

(4)

Private members:

Cname of type character array

Pnumber of type long

No_of_units of type integer

Amount of type float.

Calc_Amount() This member function should calculate the amount as $\text{No_of_units} \times \text{Cost}$.

Amount can be calculated according to the following conditions:

No_of_units Cost

First 50 units Free

Next 100 units 0.80 @ unit

Next 200 units 1.00 @ unit

Remaining units 1.20 @ unit

Public members:

* A function **Accept()** which allows user to enter Cname,

Pnumber, No_of_units and invoke function **Calc_Amount()**.

* A function **Display()** to display the values of all the data members on the screen.

Ans.

class Ele_Bill

{

char Cname[20];

long Pnumber;

int No_of_units;

float Amount;

void Calc_Amount();

public:

void Accept();

void Display();

};

void Ele_Bill :: Calc_Amount()

{

if(No_of_units<=50)

{

Amount=0;

}

else if(No_of_units<=150)

{

Amount=(No_of_units-50)*0.80;

}

else if(No_of_units<=350)

{

Amount=80+(No_of_units-150)*1.00;

```

    }
        else
        {
            Amount=80+200+(No_of_units-350)*1.20;
        }
    }
    void Ele_Bill :: Accept( )
    {
        gets(Cname);
        cin>Pnumber>>No_of_units;
        Calc_Amount( );
    }

    void Ele_Bill :: Display( )
    {
        cout<<Cname<<Pnumber<<No_of_units<<Amount;
    }
}

```

(d) Answer the questions (i) to (iv) based on the following:

(4)

```

class Faculty
{
    int FCode;
protected:
    char
    FName[20];
public:
    Faculty( );
    void Enter( );
    void Show( );
};

class Programme
{
    int PID;
protected:
    char Title[30];
public:
    Programme ( );
    void Commence( );
    void View( );
};

class Schedule: public Programme, Faculty
{
    int DD,MM,YYYY;
public:
    Schedule( );
    void Start( );
};

```

```

        void View( );
};
void main( )
{
    Schedule S;          //Statement 1
    _____          //Statement 2
}

```

OR

Consider the following class State :

```

class State
{
protected :
int tp;
public :
State( ) { tp=0;}
void inctp( ) { tp++;}
int gettp( ); { return tp; }
};

```

Write a code in C++ to publically derive another class 'District' with the following additional members derived in the public visibility mode. (4)

Data Members:

```

Dname      string
Distance    float
Population  long int

```

Member functions:

DINPUT() : To enter Dname, Distance and population

DOUTPUT() : To display the data members on the screen.

(i) Write the names of all the member functions, which are directly accessible by the object S of class Schedule as declared in main() function.

Ans.

Start(), Schedule::View(), Commence(), Programme::View()

(ii) Write the names of all the members, which are directly accessible by the member function Start() of class Schedule.

Ans.

DD,MM,YYYY, Schedule::View()
Title, Commence(), Programme::View()
Fname, Enter(), Show()

(iii) Write Statement 2 to call function View() of class Program from the object S of class Schedule.

Ans.

S.Programme::View();

(iv) What will be the order of execution of the constructors, when the object S of class Schedule is declared inside main()?

Ans. Programme(), Faculty(), Schedule()

OR

Ans.

```
class District : public State
{
public :
char    Dname[20];
float    Distance;
long int Population;
void DINPUT( )
{
gets(Dname);
cin>>distance;
cin>>Population;
}
void DOUTPUT( )
{
cout<<Dname<<endl;
cout<<Distance<<endl;
cout<<population<<endl;
}
};
```

- Names of the constructor/class without parenthesis is acceptable

Q.3.: (a) Write a user-defined function AddEnd4(int A[][4],int R,int C) in C++ to find and display the sum of all the values, which are ending with 4 (i.e., unit place is 4).

For example if the content of array is:

24	16	14
19	5	4

The output should be 42

(3)

OR

Write a user defined function in C++ to find the sum of both left and right diagonal elements from a two dimensional array.

(3)

Ans.

```
void AddEnd4(int A[ ][4], int R, int C)
{
    int I,J,sum=0;
    for(I=0;I<R;I++)
    {
        for(J=0;J<C;J++)
        if(A[I][J]%10 ==4)
            sum=sum+A[I][J];
    }
    cout<<sum;
}
```

OR

```
void Diagsumboth(int A[][4], int n)
{
    int sumLt=0,sumRt=0;
    for(int i=0;i<n;i++)
    {
        sumLt+=A[i][i];
        else
        sumRt+=A[n-1-i][i];
    }
    cout<<"sum of left
    diagonal"<<sumLt<<endl;
    cout<<"sum of right diagonal"<<sumRt<<endl;
}
```

(b) Write a user-defined function EXTRA_ELE(int A[], int B[], intN) in C++ to find and display the extra element in Array A. Array A contains all the elements of array B but one more element extra. (Restriction: array elements are not in order)

Example If the elements of Array A is 14, 21, 5, 19, 8, 4, 23, 11

and the elements of Array B is 23, 8, 19, 4, 14, 11, 5

Then output will be 21

(2)

OR

Write a user defined function Reverse(int A[],int n) which accepts an integer array and its size as arguments(parameters) and reverse the array.

Example :

if the array is 10,20,30,40,50 then reversed array is 50,40,30,20,10

Ans.

```
void EXTRA_ELE(int A[], int B[],int N)
{
    int i,j,flag=0;
    for(i=0;i<N;i++)
    {
        for(j=0;j<N;j++)
        {
```

(2)

```

        if(A[i]==B[j])
        {
            flag=1;
            break;
        }
    }
    if(flag==0)
        cout<<"Extra element"<<A[i];
flag=0;
}
}

```

OR

```

void Reverse( int A[ ], int n)
{
    int temp;
    for(int i=0;i<n/2;i++)
    {
        temp=A[i];
        A[i]=A[n-1-i];
        A[n-1-i]=temp;
    }
}

```

OR

(c) An array S[10] [30] is stored in the memory along the column with each of its element occupying 2 bytes. Find out the memory location of S[5][10], if element S[2][15] is stored at the location 8200.

(3)

OR

An array A[30][10] is stored in the memory with each element requiring 4 bytes of storage ,if the base address of A is 4500 ,Find out memory locations of A[12][8], if the content is stored along the row.

(3)

Ans.

OPTION 1:

ASSUMING LBR=LBC=0

W=2 BYTES, NUMBER OF ROWS(M)=10, NUMBER OF COLUMNS(N)=30

$LOC(S[I][J]) = B + (I + J * M) * W$

$LOC(S[2][15]) = B + (2 + 15 * 10) * 2$

$8200 = B + (152 * 2)$

$B = 8200 - 304$

$B = 7896$

$LOC(S[5][10]) = 7896 + (5 + 10 * 10) * 2$

$= 7896 + (105 * 2)$

$$= 7896 + 210$$

$$= 8106$$

OPTION 2:

ASSUMING LBR=2, LBC=15 AND B = 8200

W=2 BYTES, NUMBER OF ROWS(M)=10,

NUMBER OF

COLUMNS(N)=30

$$\text{LOC}(S[I][J]) = B + ((I - \text{LBR}) + (J - \text{LBC}) * M) * W$$

$$\text{LOC}(S[5][10]) = 8200 + ((5 - 2) + (10 - 15) * 10) * 2$$

$$= 8200 + (3 + (-5) * 10) * 2$$

$$= 8200 + (3 + (-50)) * 2$$

$$= 8200 + (3 - 50) * 2$$

$$= 8200 + (-47) * 2$$

$$= 8200 - 94$$

$$= 8106$$

OR

$$\text{Locof } A[12][8] = B + W * (N * (I - \text{LBR}) + (J - \text{LBC}))$$

$$= 4500 + 4 * (10 * 12 + 8)$$

$$= 4500 + 4 * (128)$$

$$= 4500 + 512$$

$$= 5012$$

(d) Write the definition of a member function `Ins Player()` for a class `CQUEUE` in C++, to add a `Player` in a statically allocated circular queue of `PLAYERS` considering the following code Is already written as a part of the program: (4)

```
struct Player
```

```
{
```

```
    long Pid;
```

```
    char Pname[20];
```

```
};
```

```
const int size=10;
```

```
class CQUEUE
```

```
{
```

```
    Player Ar[size];
```

```
int Front, Rear;
```

```
public:
```

```
    CQUEUE( )
```

```
{
```

```
    Front = -1;
```

```
    Rear = -1;
```

```
}
```

```
void Ins_Player();    // To add player in a static circular queue
```

```
void Del_Player();    // To remove player from a static circular queue
```

```
void Show_Player();    // To display static circular queue
```

};

OR

Write a function in C++ to delete a node containing Books information ,from a dynamically allocated stack of Books implemented with the help of the following structure:

(4)

```
struct Book
```

```
{
```

```
int BNo;
```

```
char BName[20];
```

```
Book *Next;
```

```
};
```

Ans.

```
void CQUEUE :: Ins_Player( )
```

```
{
```

```
    if((Front==0 && Rear==size-1) || (Front==Rear+1)
```

```
    {
```

```
        cout<< "Overflow";
```

```
        return;
```

```
    }
```

```
    else if(Rear == -1)
```

```
    {
```

```
        Front=0;
```

```
        Rear=0;
```

```
    }
```

```
    else if(Rear==size-1)
```

```
    {
```

```
        Rear=0;
```

```
}
```

```
    else
```

```
{
```

```
        Rear++;
```

```
}
```

```
    cout<< "Enter Player Id=";
```

```
    cin>>Ar[Rear].Pid;
```

```
    cout<< "Enter Player Name=";
```

```
    gets(Ar[Rear].Pname);
```

```
}
```

OR

```
struct Book
```

```
{
```

```
int BNo;
```

```
char BName[20];
```

```
Book *Next;
```

```
}*temp,*top;
```

```
void pop()
{
temp=new Book
;
temp=top;
top=top->next;
delete temp;
}
```

(e)

Convert the following Infix expression to its equivalent Postfix expression, showing the stack contents for each step of conversion.

A/B+C*(D

(2)

OR

Evaluate the following Postfix expression :

4,10,5,+,*,15,3,/,-

Element	Stack	Postfix
A		A
/	/	A
B	/	AB
+	+	AB/
C	+	AB/C
*	+	AB/C
(	+(	AB/C
D	+(	AB/CD
-	+(	AB/CD
E	+(	AB/CDE
)	+	AB/CDE-
	+	AB/CDE-*

55

```
void RevText( )
```

Question 4:

(a) Write a function RevText() to read a text file " Input.txt " and Print only word starting with 'I' in reverse order .

(3)

Example: If value in text

file is: INDIA IS MY COUNTRY

Output will be: AIDNI SI MY COUNTRY

OR

Write a function in C++ to count the number of lowercase alphabets present in a text file "BOOK..txt".

(3)

Ans.

```
{
    ifstream Fin("Input.txt");
    char Word[20];
    while(!Fin.eof())
    {
        Fin>>Word;
        if(Word[0]=='l')
            strrev(Word);
        cout<<Word<< " ";
    }
    Fin.close( );
}
```

OR

```
int Countalpha()
ifstream ifile ("BOOK.txt");
char ch;
int count =0;
while (! ifile.eof())
{
    ifile.get(ch);
    ifile.get(ch);
    if(islower(ch))
        count ++;
}
ifile.close();
return (count)
}
```

(b) Write a function in C++ to search and display details, whose destination is "Cochin" from binary file "Bus.Dat". Assuming the binary file is containing the objects of the following class:

(2)

class BUS

```
{    int Bno;                // Bus Number
    char From[20];          // Bus Starting Point
    char To[20];            // Bus Destination
```

public:

```
char * StartFrom ( ); { return From; }
char * EndTo( ); { return To; }
void input() { cin>>Bno>> gets(From); get(To); }
```

```
void show( ) { cout<<Bno<< ":"<<From << ":" <<To<<endl; }
};
```

OR

Write a function in C++ to add more new objects at the bottom of a binary file "STUDENT.dat", assuming the binary file is containing the objects of the following class :

(2)

```
class STU
{
int Rno;
char Sname[20];
public: void Enter( )
{
cin>>Rno;gets(Sname);
}
void show( )
{
count << Rno<<sname<<endl;
}
};
```

Ans.

```
void Read_File( )
{
BUS B;
ifstream Fin;
Fin.open("Bus.Dat", ios::binary);
while(Fin.read
((char *) &B, sizeof(B)))
{
if(strcmp(B.EndTo(), "Cochin")==0)
{
B.show( ) ;
}
}
Fin.close( );
}
```

OR

```
void Addrecord()
{
ofstream ofile;
ofile.open("STUDENT.dat", ios::out);
STU S;
char ch='Y';
while (Ch=='Y' || Ch == 'y')
```

```
{
S.Enter( );
ofile.write (Char*) & S, sizeof(s));
cout << "more (Y/N)";
cin>>ch;
}
ofile.close( );
}
```

(c) Find the output of the following C++ code considering that the binary file PRODUCT.DAT exists on the hard disk with a list of data of 500 products.

(1)

```
class PRODUCT
{
    int PCode;char PName[20];
public:
    void Entry();void Disp( );
};
void main()
{
    fstream In;
    In.open("PRODUCT.DAT",ios::binary|ios::in);
    PRODUCT P;
    In.seekg(0,ios::end);
    cout<<"Total Count: "<<In.tellg()/sizeof(P)<<endl;
    In.seekg(70*sizeof(P));
    In.read((char*)&P, sizeof(P));
    In.read((char*)&P, sizeof(P));
    cout<<"At Product:"<<In.tellg()/sizeof(P) + 1;
    In.close();
}
```

OR

Which file stream is required for seekg() ?

(1)

Ans.

Total Count:500

At Product: 73

OR

fstream/ ifstream

SECTION- C

Question 5:

(8)

(a)Observe the following table and answer the parts(i) and(ii) accordingly

Table:Product

Pno	Name	Qtr	Purchase Date
101	Pen	102	12-12-2011
102	Pencil	201	21-02-2013
103	Eraser	90	09-08-2010
109	Sharpener	90	31-08-2012
113	Clips	900	12-12-2011

- (i) Write the names of most appropriate columns, which can be considered as candidate keys.

Ans.

Candidate Key: Pno, Name

- (ii) What is the degree and cardinality of the above table?

Ans.

Degree:4

Cardinality:5

- (b) Write SQL queries for (i) to (iv) and find outputs for SQL queries (v) to (viii), which are based on the tables.

TRAINER

TID	TNAME	CITY	HIREDATE	SALARY
101	SUNAINA	MUMBAI	1998-10-15	90000
102	ANAMIKA	DELHI	1994-12-24	80000
103	DEEPTI	CHANDIGARD	2001-12-21	82000
104	MEENAKSHI	DELHI	2002-12-25	78000
105	RICHA	MUMBAI	1996-01-12	95000
106	MANIPRABHA	CHENNAI	2001-12-12	69000

COURSE

CID	CNAME	FEES	STARTDATE	TID
C201	AGDCA	12000	2018-07-02	101
C202	ADCA	15000	2018-07-15	103
C203	DCA	10000	2018-10-01	102
C204	DDTP	9000	2018-09-15	104
C205	DHN	20000	2018-08-01	101

C206	O LEVEL	18000	2018-07-25	105
------	---------	-------	------------	-----

(i) Display the Trainer Name, City & Salary in descending order of their Hiredate.

Ans.

SELECT TNAME, CITY, SALARY FROM TRAINER ORDER BY HIREDATE;

(ii) To display the TNAME and CITY of Trainer who joined the Institute in the month of December 2001.

Ans.

SELECT TNAME, CITY FROM TRAINER WHERE HIREDATE
BETWEEN

'2001-12-01' AND '2001-12-31';

OR

SELECT TNAME, CITY FROM TRAINER WHERE HIREDATE >= '2001-12-01' AND
HIREDATE <= '2001-12-31';

OR

SELECT TNAME, CITY FROM TRAINER WHERE HIREDATE LIKE
'2001-12%';

OR

WHERE HIREDATE >= '2001-12-01' AND HIREDATE <= '2001-12-31'

OR

WHERE HIREDATE LIKE '2001-12%'

(iii) To display TNAME, HIREDATE, CNAME, STARTDATE from tables TRAINER and COURSE of all those courses whose FEES is less than or equal to 10000.

Ans.

SELECT TNAME, HIREDATE, CNAME, STARTDATE FROM
TRAINER, COURSE WHERE TRAINER.TID=COURSE.TID AND
FEES <= 10000;

(iii) To display number of Trainers from each city.

Ans.

SELECT CITY, COUNT(*) FROM TRAINER GROUP BY CITY;

(v) SELECT TID, TNAME, FROM TRAINER WHERE CITY NOT IN('DELHI', 'MUMBAI');

Ans.

TID TNAME

103 DEEPTI

106 MANIPRABHA

(vi) SELECT DISTINCT TID FROM COURSE;

Ans.

DISTINCT TID

101

103

102

104

105

(vii) SELECT TID, COUNT(*), MIN(FEES) FROM COURSE GROUP BY
TID HAVING COUNT(*) > 1;

Ans.

TID COUNT(*) MIN(FEES)

101 2 12000

(viii) SELECT COUNT(*), SUM(FEES) FROM COURSE WHERE STARTDATE < '2018-09-15';

Ans.

COUNT(*) SUM(FEES)

4 65000

Question 6

(8)

(a) State any one Distributive Law of Boolean Algebra and Verify it using truth table.

Ans.

Distributive Law:

$A + BC = (A + B)(A + C)$

A	B	C	BC	A+BC	(A-B)	(A+C)	(A+B)(A+C)
0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0
0	1	0	0	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

OR

$A(B+C) = AB + AC$

A	B	C	B+C	A(B+C)	AB	AC	AB+AC
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0

1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

(b) Draw the Logic Circuit of the following Boolean Expression:

$$((U + V') \cdot (U + W)) \cdot (V + W')$$

Ans.



(c) Derive a Canonical SOP expression for a Boolean function $F(X,Y,Z)$ represented by the following truth table:

X	Y	Z	$F(X,Y,Z)$
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Ans.

$$F(X,Y,Z) = X'Y'Z' + X'Y'Z + XY'Z' + XYZ$$

OR

$$F(X,Y,Z) = \sum(0,1,4,7)$$

(d) Reduce the following Boolean Expression to its simplest form using K-Map:

$$F(X,Y,Z,W) = \sum(0,1,2,3,4,5,8,10,11,14)$$

Simplified Expression: $X'Z' + Y'W' + Y'Z + XZW'$

Question 7:

(10)

(a)

Arun opened his e-mail and found that his inbox was full of hundreds of unwanted mails. It took him around two hours to delete these unwanted mails and find the relevant ones in his

inbox. What may be the cause of his receiving so many unsolicited mails? What can Arun do to prevent this happening in future?

Ans.

Arun's email is spammed. This can be due to multiple promotional mails from unwanted advertisers. He must have opened and checked some of the promotional offers emails while surfing the internet.

To prevent happening this in future, Arun should create filters in his email to stop receiving these unwanted mails and never open emails from non-trusted sources.

(b) Assume that 50 employees are working in an organization. Each employee has been allotted a separate workstation to work. In this way, all computers are connected through the server and all these workstations are distributed over two floors. In each floor, all the computers are connected to a switch. Identify the type of network?

Ans.

LAN(Local Area Network)

(c) Your friend wishes to install a wireless network in his office. Explain him the difference between guided and unguided media.

Ans.

Guided media uses cables to connect computers such as RJ45 connectors, Cat5/6 cables, STP, UTP, OFC etc. while in the case of unguided media, the transmission of data is wireless using waves. The technologies used are Wi-fi, Bluetooth, infrared, radiowaves etc.

(d) Write the expanded names for the following abbreviated terms used in Networking and Communications:

(i)

CDMA (ii) HTTP (iii) XML (iv) URL

Ans.

(i) Code Division Multiple Access

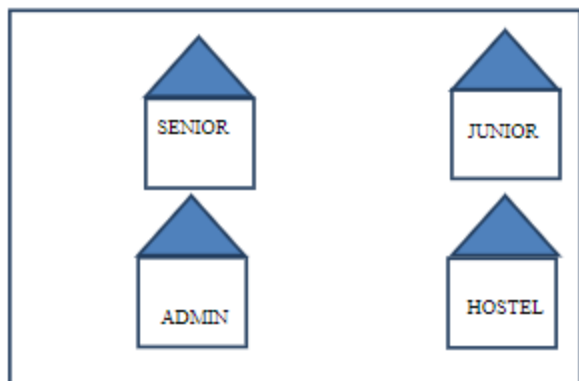
(ii) Hyper Text Transfer Protocol

(iii) Extensible Markup Language

(iv) Uniform Resource Locator

(e) Multipurpose Public School, Bangluru is Setting up the network between its Different Wings of school campus. There are 4 wings named as SENIOR(S), JUNIOR(J), ADMIN(A) and HOSTEL(H).

Multipurpose Public School, Bangluru



Distance between various wings are given below:

Wing A to Wing S	100m
Wing A to Wing J	200m
Wing A to Wing H	400m
Wing S to Wing J	300m
Wing S to Wing H	100m
Wing J to Wing H	450m

Number of computers installed at various wings are as follows:

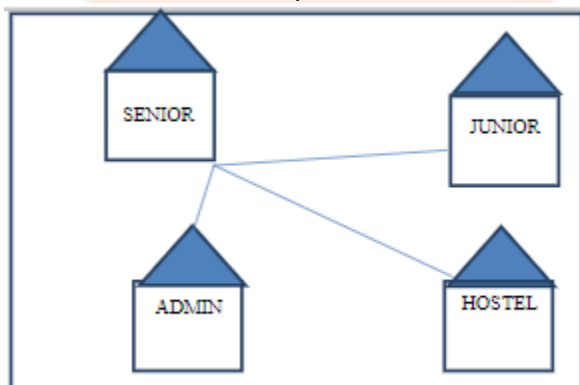
Wings	Number of computers
Wing A	20
Wing S	150
Wing J	50
Wing H	25

(i)

Suggest the best wired medium and draw the cable layout to efficiently connect various wings of Multipurpose Public School, Bangluru.

Ans:

Best wired medium: Optical Fibre / CAT5 / CAT6 / CAT7 /CAT8/ Ethernet Cable



(ii) Name the most suitable wing Where the Server should be installed. Justify your answer.

Ans.

Wing Senior(S)- since it has maximum number of computers.

(iii) Suggest a device/software and its placement that would provide data security for the entire network of the School.

Ans.

Firewall -Placed with the server at Senior

(iv)

Suggest a device and the Protocol that shall be needed to provide wireless Internet access to all smartphone/laptop users in the campus of Multipurpose Public School, Bengaluru.

Ans.

Device Name: WiFi Router OR WiMax OR RF Router OR Wireless

Modem OR RF Transmitter

Protocol : WAP OR 802.16 OR TCP/IP OR VOIP OR MACP OR 802.11